

Enterprise IoT & Embedded Automation Platform

1. Introduction

The **Enterprise IoT & Embedded Automation Platform** is an ESP32-based IoT system designed for real-time monitoring, automation, machine-condition monitoring, and alert management.

The system uses multiple sensors to collect environmental and machine-related data. The ESP32 processes this data and communicates with the Blynk IoT cloud platform through Wi-Fi. A Blynk dashboard is used to visualize sensor readings, machine status, service status, alerts, and automation states.

The project is implemented and tested using the Wokwi simulation environment.

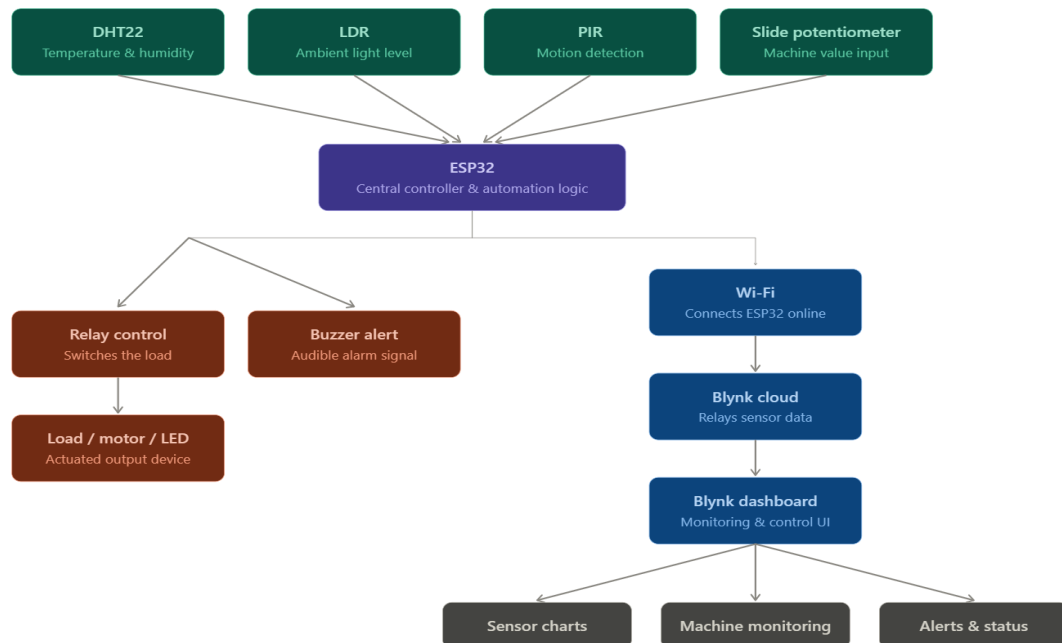
2. Objectives

The objectives of this project are:

- To develop an ESP32-based IoT monitoring system.
- To interface multiple sensors with an ESP32.
- To monitor temperature and humidity.
- To monitor ambient light and motion.
- To monitor a variable machine parameter.
- To implement automatic relay-based control.
- To provide AUTO and MANUAL operating modes.
- To connect the embedded system to Blynk IoT.
- To visualize sensor data using graphs.
- To implement warning and critical alert conditions.
- To monitor machine and service status.
- To demonstrate a basic predictive-maintenance concept.

3. System Architecture

The system consists of five major sections:



Input Section

The following inputs are provided:

- DHT22 – Temperature and Humidity
- LDR – Ambient Light
- PIR – Motion
- Slide Potentiometer – Machine Value

Processing Section

The ESP32 acts as the central controller. It reads the sensors, processes the values and evaluates predefined threshold conditions.

Decision Section

The controller determines:

- Machine condition
- Service condition
- Alert level
- Emergency status
- Operation mode

Actuation Section

The relay controls the connected load, represented by the LED/DC motor in the simulation.

Cloud Section

The processed information is transmitted to Blynk IoT and displayed on the dashboard.

4. Components Used

Component	Purpose
ESP32	Main controller and Wi-Fi communication
DHT22	Temperature and humidity measurement
LDR	Ambient light measurement
PIR Sensor	Motion detection
Slide Potentiometer	Simulated machine parameter
Relay	Load switching
LED	Load indication
Buzzer	Alert indication
Wokwi	Circuit simulation
Blynk IoT	Cloud monitoring and dashboard

5. Circuit Connections

Component	ESP32 Pin	Function
DHT22	GPIO 4	Temperature/Humidity
LDR	GPIO 35	Light level
PIR	GPIO 27	Motion
Slide Potentiometer	GPIO 34	Machine value
Relay	GPIO 26	Load control

The ESP32 receives analog and digital signals from the sensors and controls the relay based on the programmed automation logic.

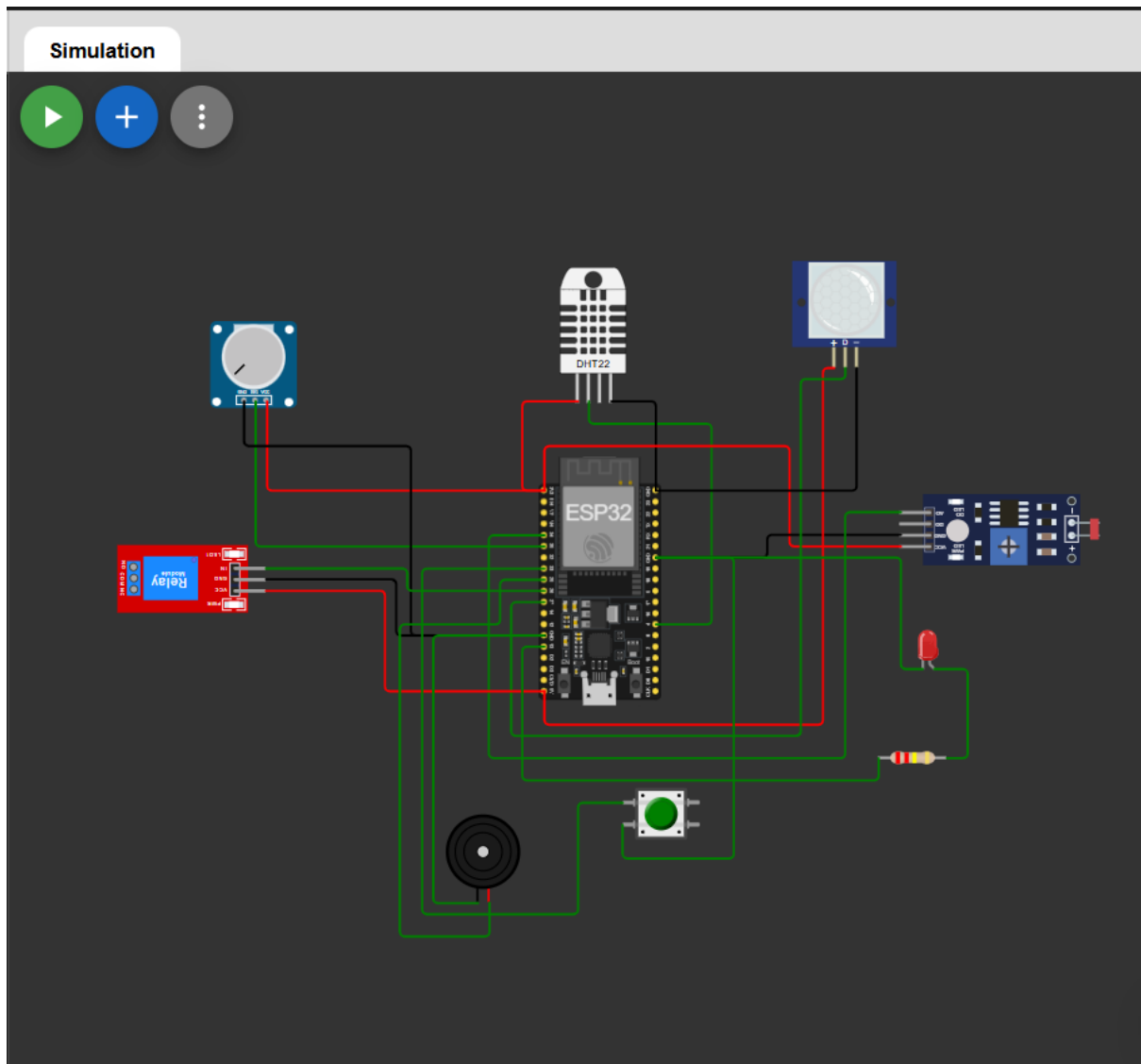


Figure 1: ESP32-based Enterprise IoT Circuit

6. Working Principle

The system starts by initializing the ESP32, sensors, relay and Blynk connection.

The DHT22 provides temperature and humidity readings. The LDR provides an analog value representing the ambient light level. The PIR sensor provides a digital motion signal. The slide potentiometer provides a variable analog value representing a machine parameter.

The ESP32 continuously evaluates these values.

If the monitored parameters are within their normal ranges, the system remains in the **NORMAL** state.

If a parameter crosses a warning threshold, the system generates a **WARNING** condition.

If a parameter reaches a critical threshold, the system generates a **CRITICAL** condition and indicates that maintenance or immediate attention is required.

7. Blynk IoT Configuration

The Blynk dashboard is used to monitor the system remotely.

Sensor Datastreams

Virtual Pin Datastream		Type	Range	Unit
V0	Temperature	Double	0–100	°C
V1	Humidity	Double	0–100	%
V2	Ambient Light	Integer	0–4095	lx
V3	Motion Status	Integer	0–1	—
V4	Machine Value	Integer	0–4095	—
V5	Machine Status	String	—	—
V6	Service Status	String	—	—
V7	Relay Status	Integer	0–1	—
V8	Load Status	Integer	0–1	—
V9	Operation Mode	String	—	—
V10	Alert Message	String	—	—
V11	Alert Level	Integer	0–2	—
V12	Emergency Status	Integer	0–1	—

8. Dashboard Design

The Blynk dashboard is divided into four sections.

8.1 Environmental Monitoring

The dashboard displays:

- Temperature
- Humidity
- Ambient Light
- Motion

The corresponding historical charts are used to observe changes over time.

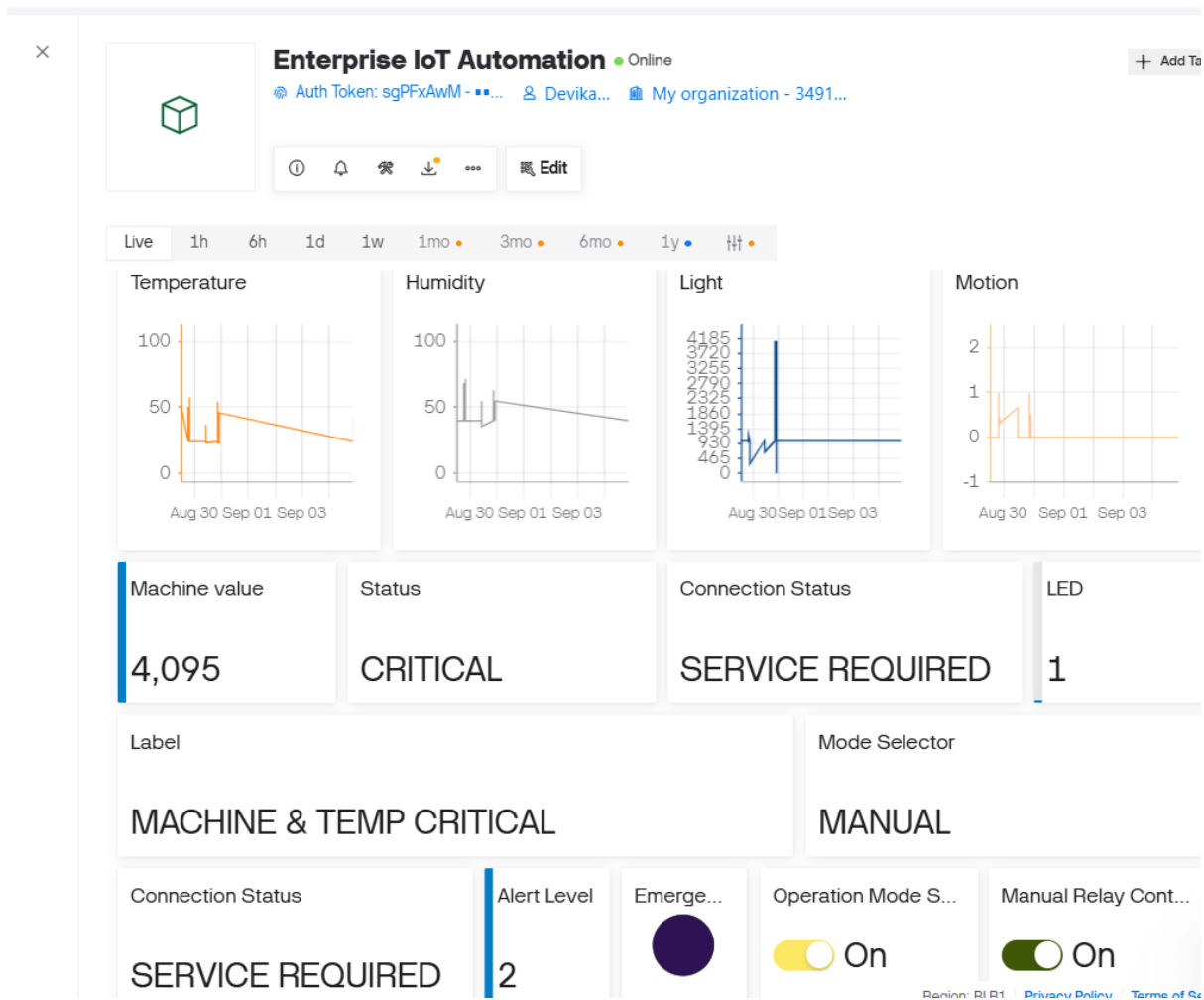


Figure 2: Environmental Monitoring Dashboard

8.2 Industrial Monitoring

This section displays:

- Machine Value
- Machine Status
- Service Status

The machine value changes according to the slide potentiometer input in Wokwi.

8.3 Automation and Control

This section displays:

- Relay Status
- Load Status
- Operation Mode

The system supports both automatic and manual operation.

8.4 Alert Monitoring

The alert section displays:

- Alert Message
- Alert Level
- Emergency Status

Alert levels are represented as:

Alert Level	Value	Meaning
NORMAL	0	System operating normally
WARNING	1	Parameter requires attention
CRITICAL	2	Immediate attention/maintenance required

9. Automation Logic

The system supports two operation modes.

AUTO Mode

In AUTO mode, the relay is controlled automatically.

The basic condition is:

Low light + Motion detected → Relay ON

Otherwise:

Relay OFF

This demonstrates automatic lighting/load control based on environmental conditions.

MANUAL Mode

In MANUAL mode, the user can control the relay manually through the Blynk dashboard.

This provides manual override functionality.

10. Temperature Monitoring and Alert Conditions

The system uses two temperature thresholds.

Normal Condition

Temperature < 35°C

The system remains normal if there are no other abnormal conditions.

Warning Condition

Temperature $\geq 35^{\circ}\text{C}$

The system generates:

- Alert Message: TEMPERATURE WARNING
- Alert Level: WARNING

Critical Condition

Temperature $\geq 40^{\circ}\text{C}$

The system generates:

- Alert Message: HIGH TEMPERATURE
- Alert Level: CRITICAL
- Service Status: TEMPERATURE CHECK REQUIRED

11. Machine Monitoring and Alert Conditions

The slide potentiometer is used to simulate a machine parameter.

The ESP32 ADC provides a value between approximately **0 and 4095**.

The machine value is compared with predefined warning and critical thresholds.

Normal

The machine value is below the warning threshold.

Machine State: NORMAL

Service Status: SERVICE OK

Alert Message: SYSTEM NORMAL

Warning

When the machine value reaches the warning threshold:

Machine State: WARNING

Service Status: INSPECTION REQUIRED

Alert Message: CHECK MACHINE

Alert Level: WARNING

Critical

When the machine value reaches the critical threshold:

Machine State: CRITICAL

Service Status: MAINTENANCE REQUIRED

Alert Message: MACHINE FAULT

Alert Level: CRITICAL

12. Emergency Status

Emergency status is represented using a binary value.

Value Status

0 Emergency inactive

1 Emergency active

The status is transmitted through **V12 – Emergency Status**.

13. Predictive Maintenance Concept

The machine-monitoring section provides a basic foundation for predictive maintenance.

The machine parameter is continuously monitored and classified into:

NORMAL → WARNING → CRITICAL

This approach can be expanded in a real industrial implementation using:

- Historical machine data
- Vibration analysis
- Temperature trends
- Current monitoring
- Machine-learning algorithms
- Anomaly detection
- Failure prediction

The present project demonstrates the initial threshold-based stage of predictive maintenance.

14. Testing

The system was tested using the Wokwi simulation environment.

Test Case 1 – Temperature Normal

Input: Temperature below 35°C

Expected:

Alert Level = NORMAL

Test Case 2 – Temperature Warning

Input: Temperature between 35°C and 39.99°C

Expected:

Alert Level = WARNING

Alert Message = TEMPERATURE WARNING

Test Case 3 – Critical Temperature

Input: Temperature $\geq 40^{\circ}\text{C}$

Expected:

Alert Level = CRITICAL

Alert Message = HIGH TEMPERATURE

Test Case 4 – Motion Detection

Input: PIR detects motion

Expected:

Motion Status = DETECTED

Test Case 5 – Machine Warning

Input: Increase slide potentiometer above the warning threshold.

Expected:

Machine Status = WARNING

Service Status = INSPECTION REQUIRED

Test Case 6 – Machine Critical

Input: Increase slide potentiometer above the critical threshold.

Expected:

Machine Status = CRITICAL

Service Status = MAINTENANCE REQUIRED

Test Case 7 – Automatic Relay

Input: Low light and motion detected in AUTO mode.

Expected:

Relay = ON

Test Case 8 – Manual Control

Input: Switch to MANUAL mode and operate the relay through Blynk.

Expected:

Relay follows the manual command.

15. Results

The implemented system successfully demonstrated:

- ESP32 sensor integration
- Temperature monitoring
- Humidity monitoring
- Ambient-light monitoring
- Motion detection
- Machine-value monitoring
- Machine-condition classification
- Relay-based automation
- AUTO/MANUAL operation
- Blynk cloud connectivity
- Historical sensor charts
- Alert-level generation
- Emergency-status monitoring
- Service-status indication

The Serial Monitor was used to verify the embedded-system output, while the Blynk dashboard was used for remote visualization.

Figure 3: Serial Monitor Output

Insert Serial Monitor screenshot here.

Figure 4: Blynk Dashboard Output

Insert Blynk dashboard screenshot here.

16. Advantages

- Real-time monitoring
- Remote cloud-based visualization
- Automated load control
- Manual override
- Machine-condition monitoring
- Alert generation
- Service-status indication
- Historical graphs
- Low-cost prototyping
- Easy scalability

17. Limitations

- The current system is simulated in Wokwi.
- The potentiometer represents a simulated machine parameter.
- The alert system uses fixed thresholds.
- Advanced machine-learning-based prediction is not implemented.
- Actual power consumption is not measured.
- Physical sensor inaccuracies are not considered in simulation.

18. Future Scope

The project can be enhanced by adding:

- Real industrial sensors
- Vibration sensors
- Current and voltage monitoring
- Energy monitoring
- Machine-learning-based predictive maintenance
- SMS/email/mobile notifications
- Database storage
- Advanced analytics

- OTA firmware updates
- Physical emergency-stop system
- Battery or solar power
- Industrial communication protocols

19. Conclusion

The **Enterprise IoT & Embedded Automation Platform** successfully demonstrates the integration of embedded systems, sensors, automation and cloud-based IoT monitoring.

The ESP32 acts as the central controller and collects data from environmental and machine-monitoring inputs. The collected information is processed using threshold-based logic to determine machine condition, service status and alert level.

The Blynk IoT dashboard provides a convenient interface for monitoring the system remotely, while the relay enables automatic and manual control of the connected load.

The project therefore provides a practical foundation for developing more advanced **Industrial IoT, automation and predictive-maintenance systems**.

20. Appendix

A. Project Files

The project repository contains the complete source code, circuit design, dashboard configuration, documentation, deployment strategy, and testing evidence for the **Enterprise IoT & Embedded Automation Platform**.

Important project files include:

- README.md – Project overview, objectives, features, and setup instructions.
- src/enterprise_iot_automation.ino – Complete ESP32 source code.
- circuit/enterprise_iot_architecture.png – System architecture and circuit representation.
- screenshots/wokwi_circuit.png – Wokwi simulation circuit.
- screenshots/serial_monitor.png – ESP32 Serial Monitor output.
- screenshots/blynk_dashboard.png – Blynk IoT dashboard.
- screenshots/blynk_dashboard_lightoff.png – Dashboard state when the light/load is OFF.
- screenshots/blynk_dashboard_lighton.png – Dashboard state when the light/load is ON.
- screenshots/relay_lightoff.png – Relay and load OFF condition.
- screenshots/relay_lighton.png – Relay and load ON condition.
- screenshots/temperature.png – Simulated temperature monitoring.

- screenshots/humidity.png – Simulated humidity monitoring.
- screenshots/motion.png – Motion/PIR sensor monitoring.
- screenshots/machine_value.png – Machine parameter monitoring.
- screenshots/alert_status.png – Alert and emergency status indication.

These files provide the implementation and visual evidence required to reproduce and verify the project.

B. Source Code

The complete **ESP32-based source code** used for the Enterprise IoT & Embedded Automation Platform is maintained in the project repository under the src directory.

The source code integrates:

- DHT22 temperature and humidity sensing
- LDR-based ambient light monitoring
- PIR-based motion detection
- Machine-value monitoring using a potentiometer
- Relay-based load control
- Automatic and manual operating modes
- Blynk IoT cloud connectivity
- Real-time parameter monitoring
- Alert-level and emergency-status handling
- Serial Monitor debugging and status reporting

The program also implements threshold-based monitoring for temperature and machine values and automatically controls the relay according to the selected operating mode.

C. Screenshots

The screenshots are maintained separately as part of the assignment submission. They provide visual evidence of the system implementation, simulation, cloud dashboard, automation, and testing results.

The screenshots include:

- Wokwi circuit configuration
- ESP32 Serial Monitor output
- Blynk IoT dashboard
- Temperature and humidity readings
- Ambient light readings

- Motion detection status
- Machine-value readings
- Relay/load ON and OFF conditions
- AUTO and MANUAL operating modes
- Alert-level indication
- Emergency-status indication

These screenshots demonstrate that the simulated embedded system communicates with the Blynk platform and that the configured monitoring and automation functions operate as intended.

D. Wokwi Project:

(<https://wokwi.com/projects/472225709350528001>)